

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 One right-angled triangle

1. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? **CBSE 2025, 1 mark**

SOLUTION

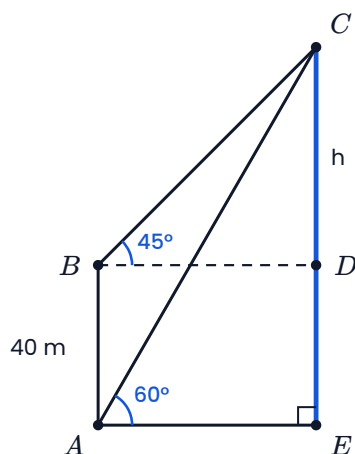
The height is the side opposite the 30° angle and the string is the hypotenuse.

$$\sin 30^\circ = \frac{150}{\text{string}}, \text{ so string} = \frac{150}{\frac{1}{2}}$$

Answer: 300 m

2 Two right-angled triangles

2. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60° . From a deck at B , 40 m directly above A , the angle of elevation of C is 45° . D is the point on CE level with B , and $CD = h$ metres. (i) Write BD in terms of h . (ii) Write BC in terms of h . (iii) Find the height CE , taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m. **CBSE 2025, 4 marks**



SOLUTION

(i) In $\triangle BDC$, $\tan 45^\circ = \frac{CD}{BD}$, so $BD = CD$. $BD = h$ metres.

(ii) $\sin 45^\circ = \frac{CD}{BC}$, so $BC = \frac{h}{\frac{1}{\sqrt{2}}}$. $BC = \sqrt{2}h$ metres.

(iii) $AE = BD = h$ and $CE = h + 40$. In $\triangle AEC$, $\tan 60^\circ = \frac{h + 40}{h}$.

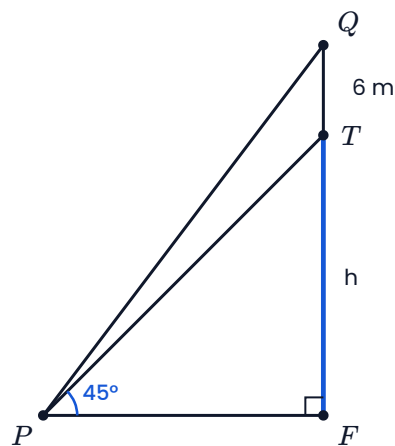
$$\sqrt{3}h = h + 40, \text{ so } h = \frac{40}{\sqrt{3} - 1} = \frac{40(\sqrt{3} + 1)}{2} = 20(\sqrt{3} + 1) = 20 \times 2.73 = 54.6$$

$CE = 54.6 + 40$. **The lighthouse is 94.6 m high.**

OR: in $\triangle AEC$, $\cos 60^\circ = \frac{AE}{AC}$, so $AE = 100 \times \frac{1}{2}$. $AE = 50$ m.

3. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

CBSE 2024, 5 marks



FT is the tower and TQ is the pole

SOLUTION

Let the height of the tower be h m and the distance of P from its foot be d m.

The angle of depression of P from the top of the tower is 45° , so the angle of elevation of the top of the tower from P is 45° : $\tan 45^\circ = \frac{h}{d}$, so $d = h$.

For the top of the pole: $\tan 60^\circ = \frac{h+6}{d}$, so $\sqrt{3}h = h+6$.

$$h = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1) = 3 \times 2.73 = 8.19$$

Answer: the tower is 8.19 m high and P is 8.19 m from its foot.

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